# Mathematical Induction

- Basis → Proove statement for $n = n_0$. (Needs to be Prooved)

- Hypothesis → Assume statement to be True for $n = k$. (Assumed) needs to be
  → Proove for $n = k+1$ ✓ — (Prooved)

⇒ Statement is True for all no. $(n) \geq n_0$

**Ex.** Coins → 3, 5

**Stmt** We can make any amount of 8 or more using only coins 3, 5

- Basis ($n = 8$)

Since $8 = 5 + 3$, Statement is True.

$8 = \underline{5} + 3$

$\hookrightarrow 9 = 3 + 3 + 3$ } Replace 5 with 2 3's ✓

$\hookrightarrow 10 = \underline{5} + 5$

$\hookrightarrow 11 = 3 + 3 + 5$

$\hookrightarrow 12 = 3 + 3 + 3 + 3$ } Replacing 3 3's with

$\hookrightarrow 13 = 5 + 5 + 3$ } 2 5's. ✓

* Hypothesis : It is Possible to make any amount k using Coins of 3 & 5.

* Case - I

* If Coins of 3, 5 both are Used, then by replacing 5 with 2 3's, we Can make k+1 amount.

* Case - II

- If Coins of only 3 are Used then by replacing 3 3's with 2 5's, we Can make k+1 amount.

- So, it is Prooved that any amount $\geq 8$ Can be made using Coins of 3 and 5.

Ex: $1^2 + 2^2 + \dots n^2 = \frac{n(n+1)(2n+1)}{6}$ $(n \geq 1)$

1. Basis $(n = 1)$

$L.H.S = 1^2 = 1$

$R.H.S = \frac{1(1+1)(2(1)+1)}{6} = \frac{1 \times 2 \times 3}{6} = 1$

- $L.H.S = R.H.S$

2. Hypothesis

Assume

$1^2 + 2^2 + \dots k^2 = \frac{k(k+1)(2k+1)}{6}$

$\Rightarrow$

- Using the assumption, Prove that

$\underbrace{1^2 + 2^2 + \dots (k+1)^2}_{(L.H.S)} = \underbrace{\frac{(k+1)((k+1)+1)(2(k+1)+1)}{6}}_{(R.H.S)}$

L.H.S $1^2 + 2^2 + \dots (k+1)^2$

$= \underbrace{1^2 + 2^2 + \dots + k^2} + (k+1)^2$

$= \frac{k(k+1)(2k+1)}{6} + (k+1)^2$

$= \frac{k(k+1)(2k+1) + 6(k+1)^2}{6}$

$$= (k+1)\left[\frac{k(2k+1)}{6} + 6(k+1)\right]$$

$$= (k+1)\frac{(2k^2 + 7k + 1)}{6}$$

$$= \frac{(k+1)(k+2)(2k+3)}{6}$$

$$= (k+1)\frac{(k+1+1)}{6}(2(k+1)+1)$$

$$= R.H.S$$

**Ex:**

$$* \quad \frac{1}{1\cdot 2} + \frac{1}{2\cdot 3} + \frac{1}{3\cdot 4} + \cdots \frac{1}{n(n+1)} = \frac{n}{n+1}$$

$(\text{for } n \geq 1)$

Ex. Show that $n^3 + 2n$ is divisible by 3. $(n \geq 1)$

- Basis $(n=1)$

$n^3 + 2n = (1)^3 + 2(1) = 3$ is divisible by 3

- Hypothesis : $k^3 + 2k$ is divisible by 3

Prove that $(k+1)^3 + 2(k+1)$ is divisible by 3.

$\Rightarrow (k+1)^3 + 2(k+1)$

$= k^3 + 3k^2 + 3k + 1 + 2k + 2$

$= (k^3 + 2k) + (3k^2 + 3k + 3)$

$= (k^3 + 2k) + 3(k^2 + k + 1)$

$\Rightarrow$ First Term is divisible by 3 as Per HyPothesis and Second term is of form 3m So it is also divisible by 3.

$\Rightarrow$ So, $(k+1)^3 + 2(k+1)$ is divisible by 3 #

Ex. $n^4 - 4n^2$ is divisible by 3
(for $n \geq 2$)

# Hypothesis $k^4 - 4k^2$ is divisible by 3.

$\Rightarrow (k+1)^4 - 4(k+1)^2$

$= k^4 + 4k^3 + 6k^2 + 4k + 1 - 4k^2 - 8k - 4$

$= (k^4 - 4k^2) + 4k^3 - 4k + 6k^2 - 3$

$=$ Hypothesis $\quad 3 \cdot m$

$=$

$4k(k^2 - 1)$

$= 4k(k-1)(k+1)$

$= 4 \times (k-1) \times k \times (k+1)$,

Product of 3 Consecutive No.

$1 \times 2 \times 3$

$26 \times 27 \times 28$

$48 \times 49 \times 50$

* Show that any integer Composed of $3^n$ Identical digits is divisible by $3^n$.

# Basis ($n=1$)

$3^1$ Identical digits = Ex. $222 = 6$
111
$333 = 9$
⋮
$999 = 27$

* Hypothesis: $3^k$ Identical digits is divisible by $3^k$.

* Prove that a no. Contains $3^{k+1}$ Identical digits is divisible by $3^{k+1}$

Ex. $k=1$

No → 222 222 222

$$\underbrace{222\ 222\ 222}_{Z} = \underbrace{(222)}_{X} \times \underbrace{(1001001)}_{Y}$$

$$\Rightarrow 10^6 + 10^3 + 10^0$$

Let $x$ be a number made up of $3^{k+1}$ identical digits

$$x = \underline{\underline{y}} \times \underline{\underline{z}} \qquad 3^k \cdot 3$$

- Where $y$ is a no. made up of $3^k$ identical digits and $z$ is always of form

$$z = 10^0 + 10^{3k} + 10^{2 \cdot 3k} + \text{- - - -}$$

& sum of digits of $z$ is always divisible by 3.

So, $x$ is also divisible by $3^{k+1}$ //